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Evaluating communication with parents in pediatric difficult patient-encounters – A Systematic review protocol

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Evaluating communication with parents in pediatric difficult patient-encounters – A Systematic review protocol

Original article

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Abstract

Introduction:

Breaking bad news and dealing with difficult patient encounters is a skill that medical residents must learn during their curriculum. Many different tools are available to measure communication quality, but their development and validation processes are often missing. In this paper, we present the protocol of a systematic review aiming to identify the validated tools for measuring communication skills or communication effectiveness with parents in a pediatrics setting in general, including for difficult patient encounters.

Methods and analysis:

We will conduct our systematic review according to the methodology suggested by COSMIN and reported this paper following the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P). We will include the studies in which authors developed and or validated tools for assessing the quality of communication with families by residents, physicians during patient encounters in pediatric settings. Studies assessing communication in telemedicine and studies that uses the tool to measure a different outcome than its validation will be excluded. Our search strategy will be developed by a scientific librarian and validated using the PRESS tool. Two reviewers will independently screen the studies for selection, extract data of the ones included, and assess their level of risk of bias using the COSMIN Risk of Bias checklist. We will perform a narrative synthesis on the study selection process, the characteristics of studies and, study population, the characteristics of tools identified, their process of development and or validation, and their psychometric properties. If sufficient data available, we will do quantitative analyses for each psychometric property.

Ethics and dissemination:

No ethic approbation is necessary, as there is no primary data collection. Our findings will be disseminated through a peer-review publication and at local, national and international conferences.

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Strengths and limitations

- This systematic review will provide not only a list of tools for communication with parents in pediatric difficult patient-encounters, but also, their validation proofs. These findings would become a powerful and useful tool for enhancing the pediatric difficult patient-encounters.
- In addition to the standard methodology required for a systematic review,
    - our literature search strategy will be revised by a second scientific librarian using the PRESS tool.
    - Appropriate guidDPelines, COSMIN, will be used for conducting the present systematic review.

- As limitations, the grey literature will not be explored; and Interviews taking place in a telemedicine setting will not be considered for inclusion.

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Introduction

As medical technologies evolve at an accelerating rate, it would be easy to ignore or forget some of the historically fundamental pillars of the medical profession. Among those bases, the patient-physician interaction is a key aspect that requires the continued development of skills, and this appears to be overlooked in some medical faculties [1-3]. To many physicians and residents, difficult patient encounters is a source of anxiety and they find their approach leaves room for improvement [3, 4]. In pediatrics, communication can be especially challenging because of the patient-parent-physician triad that is present at almost every consultation [5, 6]. Moreover, the way bad news is delivered can influence subsequent patient-parent-physician interactions and the parents' perception, coping capability and acceptance of the disease, as well as affect patient outcomes [3, 5, 7, 8]. Not a lot of literature exists regarding communication between physician and pediatric patients as the child's opinion regarding his doctor's communication skills is less reliable and harder to evaluate with a validated tool [9]. In contrast, many studies focus interest in communication with the patient's parents. As is with many skills, a physician's ability to effectively interact with families must be learned and improved during medical formation [1, 5, 8, 10, 11].

There are many aspects to consider when addressing difficult patient encounters. The main aspects often cited are those relating to the patient and parent's reality, those relating to the disease of the patient, those relating to the medical team's reality and those relating to the environment where the encounter takes place. It seems obvious to state that parental anxiety, understanding of the situation and personality will influence the parent-patient-physician relation. Moreover, a complicated medical situation itself (in terms of severity of illness, prognosis, treatment options) will influence the encounter and add tension to the physician's relation with parents. The physician's experience, personality and workload will also obviously influence the encounter. Environmental factors, such as the healthcare system's organization and the hospital or clinic's settings can influence the way news will be received by a parent or given by the physician [6]. Those are only some examples of the multitude of factors that can influence difficult patient and parent encounters.

Many medical faculties have taken a recent interest in improving their curriculum's ability to learn communication skills in difficult patient encounters and there are a growing number of tools in literature that are offered to help that achievement by residents. Many residents report high levels of anxiety and low levels of confidence in their communication skills for breaking bad news [4]. In the objectives of training of the Royal College of Physicians and Surgeons of Canada, some emphasis is made on the importance of good communication as an acquired skill [12]. There is currently a reform in Canada in residency to include a competency by design approach to learning, and residents will be specifically evaluated on their communication skills, although no official curriculum is currently available for pediatric residents.

Communication's quality perception is somewhat subjective as many aspects cannot be quantified, and every patient's needs are different [7]. In that context, developing an assessment tool that could accurately measure communication skills or effectiveness of such is a challenging task. Part of the challenge lies in the fact that the assessment tool should take into account the clinical setting. For

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example, delivering bad news in the emergency ward is not the same as doing so in a palliative care unit. Many tools have been developed to evaluate communication in different clinical settings, but there appears to be a lack of validation for these tools [13]. To our knowledge, no comprehensive and systematic review has taken interest in identifying the validated tools for measuring communication skills or communication effectiveness with parents in a pediatrics setting in general, and neither for difficult patient encounters. Identifying which validated tools exist and have been tested in pediatrics settings would prove very useful to better assess resident's needs regarding their learning of key communication skills for parent-physician interactions and, eventually, better address these needs. We took specific interest in difficult patient encounters, because we strongly felt that it is in these situations that communication is more challenging and would benefit the most from better expertise and related training.

Objectives

The present review aims to identify the validated tools for measuring communication skills or communication effectiveness with parents in a pediatric setting in general, including for difficult patient encounters.

Methods

We will conduct a systematic review according to the methodology suggested in COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) guidelines [14-16]. We reported the protocol of our review according to the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P) [17]. This protocol was registered on PROSPERO (CRD42020151642).

Eligibility criteria

We defined the eligibility criteria of studies using four key elements recommended in the COSMIN Methodology: 1) the construct; 2) the population(s); 3) the type of instrument(s); and 4) the measurement properties of interest. We will also consider study designs and the settings in which the studies have to be conducted.

- The construct of interest will be the evaluation of communication's quality in difficult pediatric doctor-parent encounters. We will include doctor-parent encounters after which review of the interaction will be evaluated. Many aspects of communication can be evaluated: the words used, the pace of the interview, the body language are some important aspects we think will identified in this review. To be included, the authors will have to present a tool that evaluates communication between a physician and the family of a pediatric patient.
- The target population will be doctors of any medical field interacting with pediatric patient's parents or guardians. We believe that a tool evaluating a certified doctor's communication would be applicable to a resident as well and therefore, we will not limit

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ourselves to studies focused on medical residents. Studies focusing solely on communication between patients and other medical professionals will be excluded. Evaluation of medical student's communication with parents will be excluded, as their learning objectives and expectations differ from those of a resident. If the study includes adult patients, the tool's validation analysis must be done at least with a pediatric subgroup (with a patient sample of at least 80% of participants under 16 years old). We chose 16 years old because we believe the older adolescent's parent's role in consultations is different than in younger patients.

- No restriction will be made regarding the format of instrument studied. We expect to find many different types of instruments. We expect to find some that are administered by peer review, by the parent themselves or by self-evaluation. Some will be filled in electronic format, others in paper format. In the electronic format, different means of completion will be used: fillable electronic document, application-based document, web-based document. Regarding the content of the instrument, we will only consider those whose main goal is to evaluate the construct of interest. Tools that measure general satisfaction following a hospitalization or that measure communication quality with the medical team as a whole and not just doctors specifically will not be included.
- The measurement properties of interest will be any psychometric parameter as defined in the COSMIN taxonomy [18]. Only studies reporting a tool with at least one validation proof or fidelity assessment will be included. The proofs of validation have to be reported or referenced through other sources in the studies.
- The study designs considered will be development and or validation studies. We will exclude studies in which authors used the tool for the outcome measurement without its process of development and or validation as suggested by the COSMIN guidelines [14].
- The settings considered will be anywhere physicians or residents interact with parents during a medical encounter. We will exclude interactions taking place in a telehealth context (encounters over the phone, over web platform, over videoconference, etc..).

As cited earlier, many aspects of communication must be taken into account when considering difficult patient encounters. We believe it would be impossible to address all those aspects in our search strategy. Although articles that analyzed clinical situations that did not represent an aspect of difficulty (for example, that analyzed every day communication with patient's family) were not a focus of our study, we will not exclude them during the selection process and to instead consider the setting in which the tool was used in the analysis. This will allow us to reduce the possible selection bias of having a search strategy not constructed precisely for difficult patient encounters or having to use our judgement to decide which articles present a difficult patient encounter. The only restrictive criterion regarding the setting will be that the encounter should take place in person. As previously mentioned, medical encounters via electronic messaging, phone or other indirect matter present some specific considerations and we will not be analyzing them. This protocol was first conceptualized before the COVID-19 pandemic and at that time, telemedicine saw very little use in practice, although it is now expected to gain significant importance. We believe we would find very little literature regarding the rating of communication in telemedicine in a pediatric setting, and that many criteria for measuring communication quality in face-to-face interviews will be applicable for telemedicine.

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Information sources

We will use the bibliographic databases such as Medline, Embase, Web of Science, CINAHL, PsycINFO, ERIC-EBSCO and the Cochrane Library. We chose a wide range of databases to cover comprehensive literature on what is available regarding the evaluation of communication in the pediatric setting. We will also consult the lists of bibliographic references in existing relevant reviews (if we find any), and in studies that will be included in the present review.

Search strategy

We developed the preliminary search strategy in Medline with the help of a scientific librarian. We used the following main concepts: Communication with families, Tools for assessing; Pediatric, Doctor or resident, and Measurement properties. Free and controlled vocabularies of these concepts were combined providing the first version of search strategy. The results of this latter were discussed with team members and their comments were integrated providing a new version. This latter was submitted to another scientific librarian for a revision using the PRESS peer review of electronic search strategies [19]. The final version of search strategy was approved by team members (see supplemental material). We will translate the final version of search strategy in other databases mentioned above.

No language restriction will be applied in the search strategy. If a study seems to be of interest to our review and the full text is not available in French nor English, we will be unable to read it entirely and will mention its omission in our data synthesis.

Grey literature

After consideration, grey literature will not be explored during this review. It certainly would be interesting to explore tools created that may not have gotten published but doing so systematically would require tremendous resources and time. There may be exist institutions that have created personalized tools that evaluate our construct of interest. Even if these tools have not been validated rigorously, the fact that they may have been used for a long time could serve as validation proofs. However, an environmental scan including a literature search and/or surveys [20] will be required to exhaustively identify these tools not published in peer-reviewed journals, as well as their proofs of use. Finally, we believe that most tools that have gone through a rigorous validation process were created with the ultimate objective of being published. Therefore, we have limited the present review to the tools published in peer-reviewed journals making irrelevant the grey literature.

Study records

Data management

We will export the citations identified from bibliographic databases to EndNote file. The citations will be merged, and duplicates will be removed to obtain unique citations for the selection of studies. The file will then be exported on Rayyan, a web app designed to facilitate the selection process in systematic reviews [19].

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Selection process

We will perform the process of study selection in three steps. Step 1: A pilot will independently be done by the two reviewers (OC and JM) basing on the 10% of unique citations. This pilot will permit to reviewers to have a common understanding of eligibility criteria. Step 2: After a conclusive pilot, OC and JM will independently do the selection of studies by title and abstract to exclude early non-eligible articles. Every study selected by one or both reviewers will then be considered for the next step. Step 3: OC and JM will independently select the studies retained at the Step 2 on the basis of full texts and will document the reasons of exclusion. A third person (CC) will be consulted in case of discordance or uncertainty. We will also contact through email the study authors when information is not clear or missing.

Data collection process

We will develop a codebook and a grid for our data extraction basing on the COSMIN methodology [16]. The codebook will include our variables of interest, their definitions, modalities to extract, and specific comments for particular cases. The grid including the variables and their modalities will be used for data extraction. The codebook and grid developed will be independently tested on two articles by two reviewers (OC and JM) to ensure a common understanding. After the conclusive pilot, OC and JM will independently extract the data from included articles. Disagreements observed after data extraction will be discussed by OC and JM for consensus. When the consensus is not reached, a third person will be consulted for a final decision.

Data items

We will extract data items characterizing the construct, the population, the type of instrument, and the measurement properties of interest. We will also collect information on the development and validation processes and the characteristics of studies.

Here are some of the main elements that will be extracted, when available:

- ***Study characteristics*** **-** name of first author, year of publication, setting, study design
- ***Construct characteristics*** - name, definition, content, number of items, range scores, time to complete, ease of use, if and how feedback is given to the evaluated doctor
- ***Target population characteristics*** **-** profiles, mean age in years (of the patient and his parent), number of men;
- ***Physician's characteristics*** **–** mean age in years, experience, specialty, number of men
- ***Setting of the encounter*** **–** degree of urgency, amount of people present for the interview, time of the interview, medical field concerned (emergency, hospital wards, general clinics, specialty clinics)
- ***Type of instruments*** **-** name, format, administrative type, total number of items, subjective status
- ***Development process*** **-** development steps, profiles of people involved, statistical methods used
- ***Validation process*** **-** validation steps, validation type, profiles of participants, mean age of participants, percentage of women participants, statistical methods used, etc.

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- ***Psychometric properties*** - reliability (type, name of parameter, value of parameters, p-value, etc.), validity (type, name of parameter, value of parameters, p-value, etc.), responsiveness (type, name of parameter, value of parameters, p-value, etc.), interpretability (type, name of parameter, value of parameters, p-value, etc.).

Risk of bias in individual studies

After a conclusive pilot for the assessment of study quality, OC and JM will independently evaluate the quality of included studies using the COSMIN Risk of Bias checklist [15]. A four-point rating system will be used to grade the methodological quality of each study (very good, adequate, doubtful, inadequate). Content validity, structural validity, reliability, measurement error and other pertinent elements will be evaluated with the rating system with the use of the checklist – a total of 10 boxes will be verified [15]. Each study will be graded according to its lowest-rating methodological aspect. They will discuss the discordances or uncertainties and will consult CC for the final decision if the consensus is not reached.

We will also apply the updated criteria for good measurement properties for the classification of identified tools [14]. Each of eight criteria will be rated sufficient, insufficient, and indeterminate according to the standards described in the COSMIN methodology [14].

Synthesis

We will describe the selection process using the PRISMA flowchart, and frequency counts. We will determine the concordance between reviewers at the steps 2 and 3 of study selection calculating a Cohen's kappa coefficient. The concordance will be considered sufficient if the kappa is equaled or greater than 0.70 [22].

We will perform a narrative synthesis using descriptive statistics. Characteristics of studies, construct, population, instrument, and measurement properties will be analyzed using frequency counts for dichotomous or categorical variables, and means, medians, and interquartile ranges for continuous variables. We will also analyze the data on the study quality assessment, the tool development and validation processes using graphic, and frequency counts. We will especially be interested in looking at the context, the population characteristics the tool was used in, what criteria the tool used to assess communication quality and how was the tool validated.

If we obtain sufficient data on parameters of psychometric properties, we will perform a quantitative analysis on each of them following the methods suggested in COSMIN guideline [14].

Meta-bias

Because of the nature of our review, we will not assess meta-biases. As is stated in the COSMIN methodology, publication bias is difficult to assess in systematic reviews of PROMs because of a lack of registries for studies on measurement properties [14].

Confidence in cumulative evidence

The strength of the body of evidence will be assessed using the modified grading of recommendation assessment, development and evaluation (GRADE). As proposed in the

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COSMIN methodology [16], we will use four of the five GRADE criteria including: risk of bias, i.e., the methodological quality of the included studies; inconsistency, i.e., unexplained inconsistency of results across included studies; imprecision, i.e., total sample size of the available included studies, and indirectness, i.e., evidence from different populations than the population of interest in the review [14]. The quality of the evidence will be graded for each measurement property separately. The starting point will always be the assumption that the pooled or overall measurement property is of high quality. The quality of evidence for each measurement property subsequently downgraded by one or two levels per criterion to moderate, low, or very low [14].

Ethics and dissemination

No ethic approbation is necessary, as there is no primary data collection. Our findings will be disseminated through a peer-review publication, at local, national and international conferences. It will also be disseminated the team's affiliated University for possible improvement of the curriculum. This protocol was registered with PROSPERO.

Patient and public involvement

Patients or the public were not involved in the design of this study. The results of this study will be disseminated through a peer-review publication and at local, national and international conferences, which will not necessarily be open to public. As no primary data collection is required, patient involvement's value would be small in this precise context.

Discussion:

We believe that this review's methodological quality will be high because of the different choices we made along the elaboration of this protocol. Every step of conceptualization of this review was done in accordance to the PRISMA and COSMIN guidelines [14-17]. The search strategy was developed by a team of qualified researchers and revised by an independent librarian with the PRESS review [19]. The selection process and analysis will be done rigorously by two independent reviewers. We will unfortunately not assess meta biases and search for grey literature during this review, which is a limitation.

This review was constructed during the COVID-19 pandemic and will not be addressing encounters over a telemedicine setting. We believe this literature is prone to expand greatly in the next couple of years as an increase in usage of telemedicine technologies is currently seen all around the world. It would be interesting as a subsequent study to review the literature existing the evaluation of communication quality over a telemedicine setting. This review's methodological construct and content will regardless be of use to whoever may take interest in this topic.

We believe this systematic review will serve as a portrait of the available evaluation tools with their validation proofs that measure the quality of communication in pediatric encounters. It will be available to use for physicians and researchers who want to use those tools in a trial, to educators who want to further their teaching abilities and to faculties or institutions who aim at evaluating adequately their residents' communication skills. Medical school's curriculums could benefit by having a better understanding of their resident's communication skills during their pediatric rotations, and potentially a better idea of precise aspects they need to work with a resident to

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improve in those skills. We believe this review could set guidance in which ways communication should be evaluated by the Royal College of Physicians and Surgeons of Canada, specifically in Pediatrics. Furthermore, the ultimate endpoint is giving physicians the best tools to improve their communication with families, especially during difficult patient encounters, thus improving parent satisfaction. We believe having the highest quality of interactions with families can and will improve patients' quality of care and satisfaction with clinical encounters.

Contributorship statement

Olivier Cusson: Guarantor, protocol redaction and review, prospero registration, search strategy review

Justine Mercier: Protocol review

Céline Catelin: Protocol review, search strategy review

Hervé Tchala Vignon Zomahoun: Methodology expert, protocol review, search strategy review

Nathalie Rheault: Search strategy conceptualization

Rachel Couban: PRESS review of the search strategy

Marie-Claude Battista: Initial methodology expert

Competing interests

None

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University of Sherbrooke – Department of Pediatrics

Data sharing statement

Data generated by this article will be made openly and publicly available upon publication.

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Supplemental material:

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Communication with families (Controlled vocabulary)	exp Hospital-Patient Relations/ or exp Physician-Patient Relations/	#1	72 086
Communication with families (Free text)	((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ti. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ab. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).kf.	#2	268 858
	("non technical" adj3 (abilit* or competenc* or skill?)).ti. or ("non technical" adj3 (abilit* or competenc* or skill?)).ab. or ("non technical" adj3 (abilit* or competenc* or skill?)).kf.	#3	650
	2 or 3	#4	269 475
Communication with families	1 or 4	#5	321 467
Tools for assessing (Controlled vocabulary)	exp Checklist/ or exp guideline/ or "Surveys and Questionnaires"/	#6	470 373

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Tools for assessing (Free text)	((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ti. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ab. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).kf.	#7	447 741
	("assessment method*" or "evaluation method*").ti. or ("assessment method*" or "evaluation method*").ab. or ("assessment method*" or "evaluation method*").kf.	#8	17 715
Tools for assessing (Free text)	7 or 8	#9	462 404
Tools for assessing	6 or 9	#10	844 912
Pediatric (Controlled Vocabulary)	exp adolescent/ or exp child/ or exp Hospitals, Pediatric/ or exp infant/ or exp Pediatric Emergency Medicine/ or exp Pediatric Nursing/ or exp Pediatricians/	#11	3 442 520
Pediatric (Free text)	(adolescen* or child or childhood or children or infan* or p?ediatric*).ti. or (adolescen* or child or childhood or children or infan* or p?ediatric*).ab. or (adolescen* or child or childhood or children or infan* or p?ediatric*).kf.	#12	1 860 446
Pediatric	11 or 12	#13	3 866 495
Doctor or resident (Controlled vocabulary)	exp education, graduate/ or education, medical/ or exp education, medical, graduate/ or exp education, medical, undergraduate/ or exp physicians/ or exp students, medical/ or exp students, public health/	#14	283 620
Doctor or resident (Free text)	(allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical	#15	1 265 235

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ti. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ab. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).kf.		
Doctor or resident	14 or 15	#16	1 366 539
Measurement properties (Controlled vocabulary)	exp discriminant analysis/ or exp Health Status Indicators/ or exp observer variation/ or exp "Outcome Assessment (Health Care)"/ or exp Psychometrics/ or exp "reproducibility of results"/	#17	1 655 441

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Measurement properties (Free text)	(Instrumentation or methods).fs.	#18	3 941 530
	("Comparative Study" or "Validation Studies").pt.	#19	1 915 070
	(clinimetr* or clinometr* or "outcome measure*").af.	#20	213 666
	(agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ti. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ab. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-	#21	2 263 439

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).kf.		
	(cronbach* and alpha$).ti. or (cronbach* and alpha$).ab. or (cronbach* and alpha$).kf.	#22	20 943
	(item* and (correlation* or selection* or reduction*)).ti. or (item* and (correlation* or selection* or reduction*)).ab. or (item* and (correlation* or selection* or reduction*)).kf.	#23	34 264
	((replicab* or repeated) and (measure$ or finding or result$ or test$)).ti. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).ab. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).kf.	#24	196 579
	(test and retest).ti. or (test and retest).ab. or (test and retest).kf.	#25	24 833
	18 or 19 or 20 or 21 or 22 or 23 or 24 or 25	#26	7 122 366
Measurement properties	17 or 26	#27	7 786 293
Final Total	5 and 10 and 13 and 16 and 27	#28	2 091

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	1
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	1
Support			
Sources	#5a	Indicate sources of financial or other support for the review	1
Sponsor	#5b	Provide name for the review funder and / or sponsor	1
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	1
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	8

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

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BMJ Open

Evaluating communication with parents in pediatric difficult patient-encounters – A Systematic review protocol

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Evaluating communication with parents in pediatric difficult patient-encounters – A Systematic review protocol

Original article

Registration: Prospero trial # CRD42020151642

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Abstract

Introduction:

Breaking bad news and dealing with difficult patient encounters is a skill that medical residents must learn during their curriculum. Many different tools are available to measure communication quality, but their development and validation processes are often missing. In this paper, we present the protocol of a systematic review aiming to identify the validated tools for measuring communication skills or communication effectiveness with parents in a pediatrics setting in general, including for difficult patient encounters.

Methods and analysis:

We will conduct our systematic review according to the methodology suggested by COSMIN and reported this paper following the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P). We will include the studies in which authors developed and/or validated tools for assessing the quality of communication with families by residents and/or physicians during patient encounters in pediatric settings. Studies assessing communication in telemedicine and studies that uses the tool to measure a different outcome than its validation will be excluded. Our search strategy will be developed by a scientific librarian and validated using the PRESS tool. Two reviewers will independently screen the studies for selection, extract data of the ones included, and assess their level of risk of bias using the COSMIN Risk of Bias checklist. We will perform a narrative synthesis on the study selection process, the characteristics of studies and, study population, the characteristics of tools identified, their process of development and/or validation, and their psychometric properties. If sufficient data available, we will do quantitative analyses for each psychometric property.

Ethics and dissemination:

No ethic approbation is necessary, as there is no primary data collection. Our findings will be disseminated through a peer-review publication and at local, national and international conferences.

PROSPERO registration number: CRD42020151642

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Strengths

- This systematic review will provide not only a list of tools for communication with parents in pediatric difficult patient-encounters, but also, their validation proofs.

- The search strategy will be revised by a second scientific librarian using the PRESS tool.

- Appropriate guidelines, COSMIN, will be used for conducting the present systematic review.

Limitations

- Grey literature will not be explored

- Interviews taking place in a telemedicine setting will not be explored for inclusion

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Introduction

As medical technologies evolve at an accelerating rate, it would be easy to ignore or forget some of the historically fundamental pillars of the medical profession. Among those bases, the patient-physician interaction is a key aspect that requires the continued development of skills, and this appears to be overlooked in some medical faculties [1-3]. To many physicians and residents, difficult patient encounters are a source of anxiety and they find their approach leaves room for improvement [3, 4]. In pediatrics, communication can be especially challenging because of the patient-parent-physician triad that is present at almost every consultation [5, 6]. Moreover, the way bad news is delivered can influence subsequent patient-parent-physician interactions and the parents' perception, coping capability and acceptance of the disease, as well as affect patient outcomes [3, 5, 7, 8]. Not a lot of literature exists regarding communication between physician and pediatric patients as the child's opinion regarding his doctor's communication skills is less reliable and harder to evaluate with a validated tool [9]. In contrast, many studies focus interest in communication with the patient's parents. As is with many skills, a physician's ability to effectively interact with families must be learned and improved during medical training [1, 5, 8, 10, 11].

There are many aspects to consider when addressing difficult patient encounters. The main aspects often cited are those relating to the patient and parent's reality, those relating to the disease of the patient, those relating to the medical team's reality and those relating to the environment where the encounter takes place. It seems obvious to state that parental anxiety, understanding of the situation and personality will influence the parent-patient-physician relation. Moreover, a complicated medical situation itself (in terms of severity of illness, prognosis, treatment options) will influence the encounter and add tension to the physician's relation with parents. The physician's experience, personality and workload will also obviously influence the encounter. Environmental factors, such as the healthcare system's organization and the hospital or clinic's settings can influence the way news will be received by a parent or given by the physician [6]. Those are only some examples of the multitude of factors that can influence difficult patient and parent encounters.

Many medical faculties have taken a recent interest in improving their curriculum's ability to learn communication skills in difficult patient encounters and there are a growing number of tools in literature that are offered to help that achievement by residents. Many residents report high levels of anxiety and low levels of confidence in their communication skills for breaking bad news [4]. In the objectives of training of the Royal College of Physicians and Surgeons of Canada, some emphasis is made on the importance of good communication as an acquired skill [12]. There is currently a reform in Canada in residency to include a competency by design approach to learning, and residents will be specifically evaluated on their communication skills, although no official curriculum is currently available for pediatric residents.

Communication's quality is somewhat subjective as many aspects cannot be quantified, and every patient's needs are different [7]. In that context, developing an assessment tool that could accurately measure communication skills or effectiveness of such is a challenging task. Part of the challenge lies in the fact that the assessment tool should take into account the clinical setting. For

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example, delivering bad news in the emergency ward is not the same as doing so in a palliative care unit. Many tools have been developed to evaluate communication in different clinical settings, but there appears to be a lack of validation for these tools [13]. To our knowledge, no comprehensive and systematic review has taken interest in identifying the validated tools for measuring communication skills or communication effectiveness with parents in a pediatrics setting in general, and neither for difficult patient encounters. Identifying which validated tools exist and have been tested in pediatrics settings would prove very useful to better assess resident's needs regarding their learning of key communication skills for parent-physician interactions and, eventually, better address these needs. We took specific interest in difficult patient encounters, because we strongly felt that it is in these situations that communication is more challenging and would benefit the most from better expertise and related training.

Objectives

The present review aims to identify the validated tools for measuring communication skills or communication effectiveness with parents in a pediatric setting in general, including for difficult patient encounters.

Methods

We will conduct a systematic review according to the methodology suggested in COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) guidelines [14-16]. We reported the protocol of our review according to the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P) [17]. This protocol was registered on PROSPERO (CRD42020151642).

Eligibility criteria

We defined the eligibility criteria of studies using four key elements recommended in the COSMIN Methodology: the construct; the population(s); the type of instrument(s); and the measurement properties of interest. We will also consider study designs and the settings in which the studies have to be conducted.

1) Construct

The construct of interest will be the evaluation of communication's quality in pediatric doctor-parent encounters. We will include doctor-parent encounters after which review of the interaction will be evaluated. Many aspects of communication can be evaluated: the words used, the pace of the interview, the body language are some important aspects we think will identified in this review. To be included, the authors will have to present a tool that evaluates communication between a physician and the family of a pediatric patient.

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2) Population(s)

The target population will be doctors of any medical field interacting with pediatric patient's parents or guardians. We believe that a tool evaluating a certified doctor's communication would be applicable to a resident as well and therefore, we will not limit ourselves to studies focused on medical residents. Studies focusing solely on communication between patients and other medical professionals will be excluded. Evaluation of medical student's communication with parents will be excluded, as their learning objectives and expectations differ from those of a resident. If the study includes adult patients, the tool's validation analysis must be done at least with a pediatric subgroup (with a patient sample of at least 80% of participants under 16 years old). We chose 16 years old because we believe that beyond that age, the adolescent's parent is less often present during consultations and communication quality assessments should be focused on the adolescent himself and not his parent.

3) Type of instrument(s)

No restriction will be made regarding the format of instrument studied. We expect to find many different types of instruments. We expect to find some that are administered by peer review, by the parent themselves or by self-evaluation. Some will be filled in electronic format, others in paper format. In the electronic format, different means of completion will be used: fillable electronic document, application-based document, web-based document. Regarding the content of the instrument, we will only consider those whose main goal is to evaluate the construct of interest. Tools that measure general satisfaction following a hospitalization or that measure communication quality with the medical team as a whole and not just doctors specifically will not be included.

3) Measurement properties

The measurement properties of interest will be any psychometric parameter as defined in the COSMIN taxonomy [18]. Only studies reporting a tool with at least one validation proof or fidelity assessment will be included. The proofs of validation have to be reported or referenced through other sources in the studies.

5) Study designs

The study designs considered will be development and/or validation studies. We will exclude studies in which authors used the tool for the outcome measurement without its process of development and/or validation as suggested by the COSMIN guidelines [14].

6) Settings

The settings considered will be anywhere physicians or residents interact with parents during a medical encounter. We will exclude interactions taking place in a telehealth context (encounters over the phone, over web platform, over videoconference, etc..).

As cited earlier, many aspects of communication must be taken into account when considering difficult patient encounters. We believe it would be impossible to address all those aspects in our search strategy. Although articles that analyzed clinical situations that did not represent an aspect of difficulty (for example, that analyzed every day communication with patient's family) were not an initial focus of our study, we will not exclude them during the selection process and to instead consider the setting in which the tool was used in the analysis. This will allow us to reduce the

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possible selection bias of having a search strategy not constructed precisely for difficult patient encounters or having to use our judgement to decide which articles present a difficult patient encounter. The only restrictive criterion regarding the setting will be that the encounter should take place in person. As previously mentioned, medical encounters via electronic messaging, phone or other indirect matter present some specific considerations and we will not be analyzing them. This protocol was first conceptualized before the COVID-19 pandemic and at that time, telemedicine saw very little use in practice, although it is now expected to gain significant importance. We believe we would find very little literature regarding the rating of communication in telemedicine in a pediatric setting, and that many criteria for measuring communication quality in face-to-face interviews will be applicable for telemedicine.

Information sources

We will use the bibliographic databases such as Medline, Embase, Web of Science, CINAHL, PsycINFO, ERIC-EBSCO and the Cochrane Library. We chose a wide range of databases to cover comprehensive literature on what is available regarding the evaluation of communication in the pediatric setting. We will also consult the lists of bibliographic references in existing relevant reviews (if we find any), and in studies that will be included in the present review.

Search strategy

We developed the preliminary search strategy in Medline with the help of a scientific librarian. We used the following main concepts: Communication with families, Tools for assessing; Pediatric, Doctor or resident, and Measurement properties. Free and controlled vocabularies of these concepts were combined providing the first version of search strategy. The results of this latter were discussed with team members and their comments were integrated providing a new version. This latter was submitted to another scientific librarian for a revision using the PRESS peer review of electronic search strategies [19]. The final version of search strategy was approved by team members (see supplemental material). We will translate the final version of search strategy in other databases mentioned above.

No language restriction will be applied in the search strategy. If a study seems to be of interest to our review and the full text is not available in French nor English, we will be unable to read it entirely and will mention its omission in our data synthesis.

Grey literature

After consideration, grey literature will not be explored during this review. It certainly would be interesting to explore tools created that may not have gotten published but doing so systematically would require tremendous resources and time. There may be exist institutions that have created personalized tools that evaluate our construct of interest. Even if these tools have not been validated rigorously, the fact that they may have been used for a long time could serve as validation proofs. However, an environmental scan including a literature search and/or surveys [20] will be required to exhaustively identify these tools not published in peer-reviewed journals, as well as their proofs of use. Finally, we believe that most tools that have gone through a rigorous validation process were created with the ultimate objective of being published. Therefore, we have limited

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the present review to the tools published in peer-reviewed journals making irrelevant the grey literature.

Study records

Data management

We will export the citations identified from bibliographic databases to EndNote file. The citations will be merged, and duplicates will be removed to obtain unique citations for the selection of studies. The file will then be exported on Rayyan, a web app designed to facilitate the selection process in systematic reviews [21].

Selection process

We will perform the process of study selection in three steps. Step 1: A pilot will independently be done by the two reviewers (OC and JM) basing on the 10% of unique citations. This pilot will permit to reviewers to have a common understanding of eligibility criteria. Step 2: After a conclusive pilot, OC and JM will independently do the selection of studies by title and abstract to exclude early non-eligible articles. Every study selected by one or both reviewers will then be considered for the next step. Step 3: OC and JM will independently select the studies retained at the Step 2 on the basis of full texts and will document the reasons of exclusion. A third person (CC) will be consulted in case of discordance or uncertainty. We will also contact through email the study authors when information is not clear or missing.

Data collection process

We will develop a codebook and a grid for our data extraction basing on the COSMIN methodology [16]. The codebook will include our variables of interest, their definitions, modalities to extract, and specific comments for particular cases. The grid including the variables and their modalities will be used for data extraction. The codebook and grid developed will be independently tested on two articles by two reviewers (OC and JM) to ensure a common understanding. After the conclusive pilot, OC and JM will independently extract the data from included articles. Disagreements observed after data extraction will be discussed by OC and JM for consensus. When the consensus is not reached, a third person (CC) will be consulted for a final decision.

Data items

We will extract data items characterizing the construct, the population, the type of instrument, and the measurement properties of interest. We will also collect information on the development and validation processes and the characteristics of studies.

Here are some of the main elements that will be extracted, when available:

- ***Study characteristics*** **-** name of first author, year of publication, setting, study design
- ***Construct characteristics*** - name, definition, content, number of items, range scores, time to complete, ease of use, if and how feedback is given to the evaluated doctor
- ***Target population characteristics*** **-** profiles, mean age in years (of the patient and his parent), patient's age group (infant, toddler, preschooler, adolescent), male/female ratio;

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- ***Physician's characteristics*** **–** mean age in years, experience, specialty, male/female ratio
- ***Setting of the encounter*** **–** degree of urgency, amount of people present for the interview, time of the interview, medical field concerned (emergency, hospital wards, general clinics, specialty clinics)
- ***Type of instruments*** **-** name, format, administrative type, total number of items, subjective status
- ***Development process*** **-** development steps, profiles of people involved, statistical methods used
- ***Validation process*** **-** validation steps, validation type, profiles of participants, mean age of participants, percentage of women participants, statistical methods used, etc.
- ***Psychometric properties*** - reliability (type, name of parameter, value of parameters, p-value, etc.), validity (type, name of parameter, value of parameters, p-value, etc.), responsiveness (type, name of parameter, value of parameters, p-value, etc.), interpretability (type, name of parameter, value of parameters, p-value, etc.).

Risk of bias in individual studies

After a conclusive pilot for the assessment of study quality, OC and JM will independently evaluate the quality of included studies using the COSMIN Risk of Bias checklist [15]. A four-point rating system will be used to grade the methodological quality of each study (very good, adequate, doubtful, inadequate). Content validity, structural validity, reliability, measurement error and other pertinent elements will be evaluated with the rating system with the use of the checklist – a total of 10 boxes will be verified [15]. Each study will be graded according to its lowest-rating methodological aspect. They will discuss the discordances or uncertainties and will consult CC for the final decision if the consensus is not reached.

We will also apply the updated criteria for good measurement properties for the classification of identified tools [14]. Each of eight criteria will be rated sufficient, insufficient, and indeterminate according to the standards described in the COSMIN methodology [14].

Synthesis

We will describe the selection process using the PRISMA flowchart, and frequency counts. We will determine the concordance between reviewers at the steps 2 and 3 of study selection calculating a Cohen's kappa coefficient. The concordance will be considered sufficient if the kappa is equaled or greater than 0.70 [22].

We will perform a narrative synthesis using descriptive statistics. Characteristics of studies, construct, population, instrument, and measurement properties will be analyzed using frequency counts for dichotomous or categorical variables, and means, medians, and interquartile ranges for continuous variables. We will also analyze the data on the study quality assessment, the tool development and validation processes using graphic, and frequency counts. We will especially be interested in looking at the context, the population characteristics the tool was used in, including the age of the patients, what criteria the tool used to assess communication quality and how was the tool validated.

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If we obtain sufficient data on parameters of psychometric properties, we will perform a quantitative analysis on each of them following the methods suggested in COSMIN guideline [14].

Meta-bias

Because of the nature of our review, we will not assess meta-biases. As is stated in the COSMIN methodology, publication bias is difficult to assess in systematic reviews of PROMs because of a lack of registries for studies on measurement properties [14].

Confidence in cumulative evidence

The strength of the body of evidence will be assessed using the modified grading of recommendation assessment, development and evaluation (GRADE). As proposed in the COSMIN methodology [16], we will use four of the five GRADE criteria including: risk of bias, i.e., the methodological quality of the included studies; inconsistency, i.e., unexplained inconsistency of results across included studies; imprecision, i.e., total sample size of the available included studies, and indirectness, i.e., evidence from different populations than the population of interest in the review [14]. The quality of the evidence will be graded for each measurement property separately. The starting point will always be the assumption that the pooled or overall measurement property is of high quality. The quality of evidence for each measurement property subsequently downgraded by one or two levels per criterion to moderate, low, or very low [14].

Ethics and dissemination

No ethic approbation is necessary, as there is no primary data collection. Our findings will be disseminated through a peer-review publication, at local, national and international conferences. It will also be disseminated to the team's affiliated University for possible improvement of the curriculum. This protocol was registered with PROSPERO.

Patient and public involvement

Patients or the public were not involved in the design of this study. The results of this study will be disseminated through a peer-review publication and at local, national and international conferences, which will not necessarily be open to public. As no primary data collection is required, patient involvement's value would be small in this precise context.

Discussion:

We believe that this review's methodological quality will be high because of the different choices we made along the elaboration of this protocol. Every step of conceptualization of this review was done in accordance to the PRISMA and COSMIN guidelines [14-17]. The search strategy was developed by a team of qualified researchers and revised by an independent librarian with the PRESS review [19]. The selection process and analysis will be done rigorously by two independent reviewers. We will unfortunately not assess meta biases and search for grey literature during this review, which is a limitation.

This review was constructed during the COVID-19 pandemic and will not be addressing encounters over a telemedicine setting. We believe this literature is prone to expand greatly in the next couple of years as an increase in usage of telemedicine technologies is currently seen all

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around the world. It would be interesting as a subsequent study to review the literature existing the evaluation of communication quality over a telemedicine setting. This review's methodological construct and content will regardless be of use to whoever may take interest in this topic.

We believe this systematic review will serve as a portrait of the available evaluation tools with their validation proofs that measure the quality of communication in pediatric encounters. It will be available to use for physicians and researchers who want to use those tools in a trial, to educators who want to further their teaching abilities and to faculties or institutions who aim at evaluating adequately their residents' communication skills. Medical school's curriculums could benefit by having a better understanding of their resident's communication skills during their pediatric rotations, and potentially a better idea of precise aspects they need to work with a resident to improve in those skills. We believe this review could set guidance in which ways communication should be evaluated by the Royal College of Physicians and Surgeons of Canada, specifically in Pediatrics. Furthermore, the ultimate endpoint is giving physicians the best tools to improve their communication with families, especially during difficult patient encounters, thus improving parent satisfaction. We believe having the highest quality of interactions with families can and will improve patients' quality of care and satisfaction with clinical encounters.

Contributorship statement

Olivier Cusson: Guarantor, protocol redaction and review, prospero registration, search strategy review

Justine Mercier: Protocol review

Céline Catelin: Protocol review, search strategy review

Hervé Tchala Vignon Zomahoun: Methodology expert, protocol review, search strategy review

Nathalie Rheault: Search strategy conceptualization

Acknowledgement

Rachel Couban: PRESS review of the search strategy

Marie-Claude Battista: Initial methodology expert

Competing interests

None

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University of Sherbrooke – Department of Pediatrics (no grant number)

Data sharing statement

Data generated by this article will be made openly and publicly available upon publication.

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Supplemental material:

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Communication with families (Controlled vocabulary)	exp Hospital-Patient Relations/ or exp Physician-Patient Relations/	#1	72 086
Communication with families (Free text)	((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ti. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ab. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).kf.	#2	268 858
	("non technical" adj3 (abilit* or competenc* or skill?)).ti. or ("non technical" adj3 (abilit* or competenc* or skill?)).ab. or ("non technical" adj3 (abilit* or competenc* or skill?)).kf.	#3	650
	2 or 3	#4	269 475
Communication with families	1 or 4	#5	321 467
Tools for assessing (Controlled vocabulary)	exp Checklist/ or exp guideline/ or "Surveys and Questionnaires"/	#6	470 373

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Tools for assessing (Free text)	((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ti. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ab. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).kf.	#7	447 741
	("assessment method*" or "evaluation method*").ti. or ("assessment method*" or "evaluation method*").ab. or ("assessment method*" or "evaluation method*").kf.	#8	17 715
Tools for assessing (Free text)	7 or 8	#9	462 404
Tools for assessing	6 or 9	#10	844 912
Pediatric (Controlled Vocabulary)	exp adolescent/ or exp child/ or exp Hospitals, Pediatric/ or exp infant/ or exp Pediatric Emergency Medicine/ or exp Pediatric Nursing/ or exp Pediatricians/	#11	3 442 520
Pediatric (Free text)	(adolescen* or child or childhood or children or infan* or p?ediatric*).ti. or (adolescen* or child or childhood or children or infan* or p?ediatric*).ab. or (adolescen* or child or childhood or children or infan* or p?ediatric*).kf.	#12	1 860 446
Pediatric	11 or 12	#13	3 866 495
Doctor or resident (Controlled vocabulary)	exp education, graduate/ or education, medical/ or exp education, medical, graduate/ or exp education, medical, undergraduate/ or exp physicians/ or exp students, medical/ or exp students, public health/	#14	283 620
Doctor or resident (Free text)	(allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical	#15	1 265 235

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ti. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ab. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).kf.		
Doctor or resident	14 or 15	#16	1 366 539
Measurement properties (Controlled vocabulary)	exp discriminant analysis/ or exp Health Status Indicators/ or exp observer variation/ or exp "Outcome Assessment (Health Care)"/ or exp Psychometrics/ or exp "reproducibility of results"/	#17	1 655 441

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Measurement properties (Free text)	(Instrumentation or methods).fs.	#18	3 941 530
	("Comparative Study" or "Validation Studies").pt.	#19	1 915 070
	(clinimetr* or clinometr* or "outcome measure*").af.	#20	213 666
	(agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ti. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ab. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-	#21	2 263 439

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).kf.		
	(cronbach* and alpha$).ti. or (cronbach* and alpha$).ab. or (cronbach* and alpha$).kf.	#22	20 943
	(item* and (correlation* or selection* or reduction*)).ti. or (item* and (correlation* or selection* or reduction*)).ab. or (item* and (correlation* or selection* or reduction*)).kf.	#23	34 264
	((replicab* or repeated) and (measure$ or finding or result$ or test$)).ti. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).ab. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).kf.	#24	196 579
	(test and retest).ti. or (test and retest).ab. or (test and retest).kf.	#25	24 833
	18 or 19 or 20 or 21 or 22 or 23 or 24 or 25	#26	7 122 366
Measurement properties	17 or 26	#27	7 786 293
Final Total	5 and 10 and 13 and 16 and 27	#28	2 091

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	1
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	1
Support			
Sources	#5a	Indicate sources of financial or other support for the review	1
Sponsor	#5b	Provide name for the review funder and / or sponsor	1
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	1
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	8

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

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BMJ Open

Evaluating communication with parents in pediatric patient-encounters – A Systematic review protocol

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Complete List of Authors:	Cusson, Olivier; Hôpital Fleurimont, Pediatrics Mercier, Justine; University of Sherbrooke, Family Medicine Catelin, Céline; Hôpital Fleurimont, Pediatrics Zomahoun, Hervé Tchala Vignon; Knowledge Translation Component of the Quebec SPOR-SUPPORT Unit
Primary Subject Heading:	Communication
Secondary Subject Heading:	Medical education and training, Paediatrics
Keywords:	MEDICAL EDUCATION & TRAINING, PAEDIATRICS, Quality in health care < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Evaluating communication with parents in pediatric patient-encounters – A Systematic review protocol

Original article

Registration: Prospero trial # CRD42020151642

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Abstract

Introduction:

Breaking bad news and dealing with difficult patient encounters is a skill that medical residents must learn during their curriculum. Many different tools are available to measure communication quality, but their development and validation processes are often missing. In this paper, we present the protocol of a systematic review aiming to identify the validated tools for measuring communication skills or communication effectiveness with parents in a pediatrics setting in general, including for difficult patient encounters.

Methods and analysis:

We will conduct our systematic review according to the methodology suggested by COSMIN and reported this paper following the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P). We will include the studies in which authors developed and/or validated tools for assessing the quality of communication with families by residents and/or physicians during patient encounters in pediatric settings. Studies assessing communication in telemedicine and studies that uses the tool to measure a different outcome than its validation will be excluded. Our search strategy will be developed by a scientific librarian and validated using the PRESS tool. Two reviewers will independently screen the studies for selection, extract data of the ones included, and assess their level of risk of bias using the COSMIN Risk of Bias checklist. We will perform a narrative synthesis on the study selection process, the characteristics of studies and, study population, the characteristics of tools identified, their process of development and/or validation, and their psychometric properties. If sufficient data available, we will do quantitative analyses for each psychometric property.

Ethics and dissemination:

No ethic approbation is necessary, as there is no primary data collection. Our findings will be disseminated through a peer-review publication and at local, national and international conferences.

PROSPERO registration number: CRD42020151642

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Strengths

- This systematic review will provide not only a list of tools for communication with parents in pediatric difficult patient-encounters, but also, their validation proofs.

- The search strategy will be revised by a second scientific librarian using the PRESS tool.

- Appropriate guidelines, COSMIN, will be used for conducting the present systematic review.

Limitations

- Grey literature will not be explored

- Interviews taking place in a telemedicine setting will not be explored for inclusion

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Introduction

As medical technologies evolve at an accelerating rate, it would be easy to ignore or forget some of the historically fundamental pillars of the medical profession. Among those bases, the patient-physician interaction is a key aspect that requires the continued development of skills, and this appears to be overlooked in some medical faculties [1-3]. To many physicians and residents, difficult patient encounters are a source of anxiety and they find their approach leaves room for improvement [3, 4]. In pediatrics, communication can be especially challenging because of the patient-parent-physician triad that is present at almost every consultation [5, 6]. Moreover, the way bad news is delivered can influence subsequent patient-parent-physician interactions and the parents' perception, coping capability and acceptance of the disease, as well as affect patient outcomes [3, 5, 7, 8]. Not a lot of literature exists regarding communication between physician and pediatric patients as the child's opinion regarding his doctor's communication skills is less reliable and harder to evaluate with a validated tool [9]. In contrast, many studies focus interest in communication with the patient's parents. As is with many skills, a physician's ability to effectively interact with families must be learned and improved during medical training [1, 5, 8, 10, 11].

There are many aspects to consider when addressing difficult patient encounters. The main aspects often cited are those relating to the patient and parent's reality, those relating to the disease of the patient, those relating to the medical team's reality and those relating to the environment where the encounter takes place. It seems obvious to state that parental anxiety, understanding of the situation and personality will influence the parent-patient-physician relation. Moreover, a complicated medical situation itself (in terms of severity of illness, prognosis, treatment options) will influence the encounter and add tension to the physician's relation with parents. The physician's experience, personality and workload will also obviously influence the encounter. Environmental factors, such as the healthcare system's organization and the hospital or clinic's settings can influence the way news will be received by a parent or given by the physician [6]. Those are only some examples of the multitude of factors that can influence difficult patient and parent encounters.

Many medical faculties have taken a recent interest in improving their curriculum's ability to learn communication skills in difficult patient encounters and there are a growing number of tools in literature that are offered to help that achievement by residents. Many residents report high levels of anxiety and low levels of confidence in their communication skills for breaking bad news [4]. In the objectives of training of the Royal College of Physicians and Surgeons of Canada, some emphasis is made on the importance of good communication as an acquired skill [12]. There is currently a reform in Canada in residency to include a competency by design approach to learning, and residents will be specifically evaluated on their communication skills, although no official curriculum is currently available for pediatric residents.

Communication's quality is somewhat subjective as many aspects cannot be quantified, and every patient's needs are different [7]. In that context, developing an assessment tool that could accurately measure communication skills or effectiveness of such is a challenging task. Part of the challenge lies in the fact that the assessment tool should take into account the clinical setting. For

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example, delivering bad news in the emergency ward is not the same as doing so in a palliative care unit. Many tools have been developed to evaluate communication in different clinical settings, but there appears to be a lack of validation for these tools [13]. To our knowledge, no comprehensive and systematic review has taken interest in identifying the validated tools for measuring communication skills or communication effectiveness with parents in a pediatrics setting in general, and neither for difficult patient encounters. Identifying which validated tools exist and have been tested in pediatrics settings would prove very useful to better assess resident's needs regarding their learning of key communication skills for parent-physician interactions and, eventually, better address these needs. We took specific interest in difficult patient encounters, because we strongly felt that it is in these situations that communication is more challenging and would benefit the most from better expertise and related training.

Objectives

The present review aims to identify the validated tools for measuring communication skills or communication effectiveness with parents in a pediatric setting in general, including for difficult patient encounters.

Methods

We will conduct a systematic review according to the methodology suggested in COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) guidelines [14-16]. We reported the protocol of our review according to the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P) [17]. This protocol was registered on PROSPERO (CRD42020151642).

Eligibility criteria

We defined the eligibility criteria of studies using four key elements recommended in the COSMIN Methodology: the construct; the population(s); the type of instrument(s); and the measurement properties of interest. We will also consider study designs and the settings in which the studies have to be conducted.

1) Construct

The construct of interest will be the evaluation of communication's quality in pediatric doctor-parent encounters. We will include doctor-parent encounters after which review of the interaction will be evaluated. Many aspects of communication can be evaluated: the words used, the pace of the interview, the body language are some important aspects we think will identified in this review. To be included, the authors will have to present a tool that evaluates communication between a physician and the family of a pediatric patient.

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2) Population(s)

The target population will be doctors of any medical field interacting with pediatric patient's parents or guardians. We believe that a tool evaluating a certified doctor's communication would be applicable to a resident as well and therefore, we will not limit ourselves to studies focused on medical residents. Studies focusing solely on communication between patients and other medical professionals will be excluded. Evaluation of medical student's communication with parents will be excluded, as their learning objectives and expectations differ from those of a resident. If the study includes adult patients, the tool's validation analysis must be done at least with a pediatric subgroup (with a patient sample of at least 80% of participants under 16 years old). We chose 16 years old because we believe that beyond that age, the adolescent's parent is less often present during consultations and communication quality assessments should be focused on the adolescent himself and not his parent.

3) Type of instrument(s)

No restriction will be made regarding the format of instrument studied. We expect to find many different types of instruments. We expect to find some that are administered by peer review, by the parent themselves or by self-evaluation. Some will be filled in electronic format, others in paper format. In the electronic format, different means of completion will be used: fillable electronic document, application-based document, web-based document. Regarding the content of the instrument, we will only consider those whose main goal is to evaluate the construct of interest. Tools that measure general satisfaction following a hospitalization or that measure communication quality with the medical team as a whole and not just doctors specifically will not be included.

3) Measurement properties

The measurement properties of interest will be any psychometric parameter as defined in the COSMIN taxonomy [18]. Only studies reporting a tool with at least one validation proof or fidelity assessment will be included. The proofs of validation have to be reported or referenced through other sources in the studies.

5) Study designs

The study designs considered will be development and/or validation studies. We will exclude studies in which authors used the tool for the outcome measurement without its process of development and/or validation as suggested by the COSMIN guidelines [14].

6) Settings

The settings considered will be anywhere physicians or residents interact with parents during a medical encounter. We will exclude interactions taking place in a telehealth context (encounters over the phone, over web platform, over videoconference, etc..).

As cited earlier, many aspects of communication must be taken into account when considering difficult patient encounters. We believe it would be impossible to address all those aspects in our search strategy. Although articles that analyzed clinical situations that did not represent an aspect of difficulty (for example, that analyzed every day communication with patient's family) were not an initial focus of our study, we will not exclude them during the selection process and to instead consider the setting in which the tool was used in the analysis. This will allow us to reduce the

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possible selection bias of having a search strategy not constructed precisely for difficult patient encounters or having to use our judgement to decide which articles present a difficult patient encounter. The only restrictive criterion regarding the setting will be that the encounter should take place in person. As previously mentioned, medical encounters via electronic messaging, phone or other indirect matter present some specific considerations and we will not be analyzing them. This protocol was first conceptualized before the COVID-19 pandemic and at that time, telemedicine saw very little use in practice, although it is now expected to gain significant importance. We believe we would find very little literature regarding the rating of communication in telemedicine in a pediatric setting, and that many criteria for measuring communication quality in face-to-face interviews will be applicable for telemedicine.

Information sources

We will use the bibliographic databases such as Medline, Embase, Web of Science, CINAHL, PsycINFO, ERIC-EBSCO and the Cochrane Library. We chose a wide range of databases to cover comprehensive literature on what is available regarding the evaluation of communication in the pediatric setting. We will also consult the lists of bibliographic references in existing relevant reviews (if we find any), and in studies that will be included in the present review.

Search strategy

We developed the preliminary search strategy in Medline with the help of a scientific librarian. We used the following main concepts: Communication with families, Tools for assessing; Pediatric, Doctor or resident, and Measurement properties. Free and controlled vocabularies of these concepts were combined providing the first version of search strategy. The results of this latter were discussed with team members and their comments were integrated providing a new version. This latter was submitted to another scientific librarian for a revision using the PRESS peer review of electronic search strategies [19]. The final version of search strategy was approved by team members (see supplemental material). We will translate the final version of search strategy in other databases mentioned above.

No language restriction will be applied in the search strategy. If a study seems to be of interest to our review and the full text is not available in French nor English, we will be unable to read it entirely and will mention its omission in our data synthesis.

Grey literature

After consideration, grey literature will not be explored during this review. It certainly would be interesting to explore tools created that may not have gotten published but doing so systematically would require tremendous resources and time. There may be exist institutions that have created personalized tools that evaluate our construct of interest. Even if these tools have not been validated rigorously, the fact that they may have been used for a long time could serve as validation proofs. However, an environmental scan including a literature search and/or surveys [20] will be required to exhaustively identify these tools not published in peer-reviewed journals, as well as their proofs of use. Finally, we believe that most tools that have gone through a rigorous validation process were created with the ultimate objective of being published. Therefore, we have limited

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the present review to the tools published in peer-reviewed journals making irrelevant the grey literature.

Study records

Data management

We will export the citations identified from bibliographic databases to EndNote file. The citations will be merged, and duplicates will be removed to obtain unique citations for the selection of studies. The file will then be exported on Rayyan, a web app designed to facilitate the selection process in systematic reviews [21].

Selection process

We will perform the process of study selection in three steps. Step 1: A pilot will independently be done by the two reviewers (OC and JM) basing on the 10% of unique citations. This pilot will permit to reviewers to have a common understanding of eligibility criteria. Step 2: After a conclusive pilot, OC and JM will independently do the selection of studies by title and abstract to exclude early non-eligible articles. Every study selected by one or both reviewers will then be considered for the next step. Step 3: OC and JM will independently select the studies retained at the Step 2 on the basis of full texts and will document the reasons of exclusion. A third person (CC) will be consulted in case of discordance or uncertainty. We will also contact through email the study authors when information is not clear or missing.

Data collection process

We will develop a codebook and a grid for our data extraction basing on the COSMIN methodology [16]. The codebook will include our variables of interest, their definitions, modalities to extract, and specific comments for particular cases. The grid including the variables and their modalities will be used for data extraction. The codebook and grid developed will be independently tested on two articles by two reviewers (OC and JM) to ensure a common understanding. After the conclusive pilot, OC and JM will independently extract the data from included articles. Disagreements observed after data extraction will be discussed by OC and JM for consensus. When the consensus is not reached, a third person (CC) will be consulted for a final decision.

Data items

We will extract data items characterizing the construct, the population, the type of instrument, and the measurement properties of interest. We will also collect information on the development and validation processes and the characteristics of studies.

Here are some of the main elements that will be extracted, when available:

- ***Study characteristics*** **-** name of first author, year of publication, setting, study design
- ***Construct characteristics*** - name, definition, content, number of items, range scores, time to complete, ease of use, if and how feedback is given to the evaluated doctor
- ***Target population characteristics*** **-** profiles, mean age in years (of the patient and his parent), patient's age group (infant, toddler, preschooler, adolescent), male/female ratio;

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- ***Physician's characteristics*** **–** mean age in years, experience, specialty, male/female ratio
- ***Setting of the encounter*** **–** degree of urgency, amount of people present for the interview, time of the interview, medical field concerned (emergency, hospital wards, general clinics, specialty clinics)
- ***Type of instruments*** **-** name, format, administrative type, total number of items, subjective status
- ***Development process*** **-** development steps, profiles of people involved, statistical methods used
- ***Validation process*** **-** validation steps, validation type, profiles of participants, mean age of participants, percentage of women participants, statistical methods used, etc.
- ***Psychometric properties*** - reliability (type, name of parameter, value of parameters, p-value, etc.), validity (type, name of parameter, value of parameters, p-value, etc.), responsiveness (type, name of parameter, value of parameters, p-value, etc.), interpretability (type, name of parameter, value of parameters, p-value, etc.).

Risk of bias in individual studies

After a conclusive pilot for the assessment of study quality, OC and JM will independently evaluate the quality of included studies using the COSMIN Risk of Bias checklist [15]. A four-point rating system will be used to grade the methodological quality of each study (very good, adequate, doubtful, inadequate). Content validity, structural validity, reliability, measurement error and other pertinent elements will be evaluated with the rating system with the use of the checklist – a total of 10 boxes will be verified [15]. Each study will be graded according to its lowest-rating methodological aspect. They will discuss the discordances or uncertainties and will consult CC for the final decision if the consensus is not reached.

We will also apply the updated criteria for good measurement properties for the classification of identified tools [14]. Each of eight criteria will be rated sufficient, insufficient, and indeterminate according to the standards described in the COSMIN methodology [14].

Synthesis

We will describe the selection process using the PRISMA flowchart, and frequency counts. We will determine the concordance between reviewers at the steps 2 and 3 of study selection calculating a Cohen's kappa coefficient. The concordance will be considered sufficient if the kappa is equaled or greater than 0.70 [22].

We will perform a narrative synthesis using descriptive statistics. Characteristics of studies, construct, population, instrument, and measurement properties will be analyzed using frequency counts for dichotomous or categorical variables, and means, medians, and interquartile ranges for continuous variables. We will also analyze the data on the study quality assessment, the tool development and validation processes using graphic, and frequency counts. We will especially be interested in looking at the context, the population characteristics the tool was used in, including the age of the patients, what criteria the tool used to assess communication quality and how was the tool validated.

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If we obtain sufficient data on parameters of psychometric properties, we will perform a quantitative analysis on each of them following the methods suggested in COSMIN guideline [14].

Meta-bias

Because of the nature of our review, we will not assess meta-biases. As is stated in the COSMIN methodology, publication bias is difficult to assess in systematic reviews of PROMs because of a lack of registries for studies on measurement properties [14].

Confidence in cumulative evidence

The strength of the body of evidence will be assessed using the modified grading of recommendation assessment, development and evaluation (GRADE). As proposed in the COSMIN methodology [16], we will use four of the five GRADE criteria including: risk of bias, i.e., the methodological quality of the included studies; inconsistency, i.e., unexplained inconsistency of results across included studies; imprecision, i.e., total sample size of the available included studies, and indirectness, i.e., evidence from different populations than the population of interest in the review [14]. The quality of the evidence will be graded for each measurement property separately. The starting point will always be the assumption that the pooled or overall measurement property is of high quality. The quality of evidence for each measurement property subsequently downgraded by one or two levels per criterion to moderate, low, or very low [14].

Ethics and dissemination

No ethic approbation is necessary, as there is no primary data collection. Our findings will be disseminated through a peer-review publication, at local, national and international conferences. It will also be disseminated to the team's affiliated University for possible improvement of the curriculum. This protocol was registered with PROSPERO.

Patient and public involvement

Patients or the public were not involved in the design of this study. The results of this study will be disseminated through a peer-review publication and at local, national and international conferences, which will not necessarily be open to public. As no primary data collection is required, patient involvement's value would be small in this precise context.

Discussion:

We believe that this review's methodological quality will be high because of the different choices we made along the elaboration of this protocol. Every step of conceptualization of this review was done in accordance to the PRISMA and COSMIN guidelines [14-17]. The search strategy was developed by a team of qualified researchers and revised by an independent librarian with the PRESS review [19]. The selection process and analysis will be done rigorously by two independent reviewers. We will unfortunately not assess meta biases and search for grey literature during this review, which is a limitation.

This review was constructed during the COVID-19 pandemic and will not be addressing encounters over a telemedicine setting. We believe this literature is prone to expand greatly in the next couple of years as an increase in usage of telemedicine technologies is currently seen all

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around the world. It would be interesting as a subsequent study to review the literature existing the evaluation of communication quality over a telemedicine setting. This review's methodological construct and content will regardless be of use to whoever may take interest in this topic.

We believe this systematic review will serve as a portrait of the available evaluation tools with their validation proofs that measure the quality of communication in pediatric encounters. It will be available to use for physicians and researchers who want to use those tools in a trial, to educators who want to further their teaching abilities and to faculties or institutions who aim at evaluating adequately their residents' communication skills. Medical school's curriculums could benefit by having a better understanding of their resident's communication skills during their pediatric rotations, and potentially a better idea of precise aspects they need to work with a resident to improve in those skills. We believe this review could set guidance in which ways communication should be evaluated by the Royal College of Physicians and Surgeons of Canada, specifically in Pediatrics. Furthermore, the ultimate endpoint is giving physicians the best tools to improve their communication with families, especially during difficult patient encounters, thus improving parent satisfaction. We believe having the highest quality of interactions with families can and will improve patients' quality of care and satisfaction with clinical encounters.

Contributorship statement

Olivier Cusson (garantator): Substantial contribution to all stages of the review, approved the final version and agrees to be accountable for its content.

Justine Mercier: Substantial contribution to the interpretation of the data for the review, revision of all its versions, approval of the final version and agrees to be accountable for its content.

Céline Catelin: Substantial contribution to the conception of the review, revision of all versions of the protocol, approval of the final version and agrees to be accountable for its content.

Hervé Tchala Vignon Zomahoun: Substantial contribution to the design of the review, revision of all its versions, approval of the final version and agrees to be accountable for its content.

Acknowledgements

Nathalie Rheault: Search strategy conceptualization

Rachel Couban: PRESS review of the search strategy

Marie-Claude Battista: Initial methodology expert

Competing interests

None

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Data sharing statement

Data generated by this article will be made openly and publicly available upon publication.

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Supplemental material:

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Communication with families (Controlled vocabulary)	exp Hospital-Patient Relations/ or exp Physician-Patient Relations/	#1	72 086
Communication with families (Free text)	((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ti. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ab. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).kf.	#2	268 858
	("non technical" adj3 (abilit* or competenc* or skill?)).ti. or ("non technical" adj3 (abilit* or competenc* or skill?)).ab. or ("non technical" adj3 (abilit* or competenc* or skill?)).kf.	#3	650
	2 or 3	#4	269 475
Communication with families	1 or 4	#5	321 467
Tools for assessing (Controlled vocabulary)	exp Checklist/ or exp guideline/ or "Surveys and Questionnaires"/	#6	470 373

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Tools for assessing (Free text)	((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ti. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ab. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).kf.	#7	447 741
	("assessment method*" or "evaluation method*").ti. or ("assessment method*" or "evaluation method*").ab. or ("assessment method*" or "evaluation method*").kf.	#8	17 715
Tools for assessing (Free text)	7 or 8	#9	462 404
Tools for assessing	6 or 9	#10	844 912
Pediatric (Controlled Vocabulary)	exp adolescent/ or exp child/ or exp Hospitals, Pediatric/ or exp infant/ or exp Pediatric Emergency Medicine/ or exp Pediatric Nursing/ or exp Pediatricians/	#11	3 442 520
Pediatric (Free text)	(adolescen* or child or childhood or children or infan* or p?ediatric*).ti. or (adolescen* or child or childhood or children or infan* or p?ediatric*).ab. or (adolescen* or child or childhood or children or infan* or p?ediatric*).kf.	#12	1 860 446
Pediatric	11 or 12	#13	3 866 495
Doctor or resident (Controlled vocabulary)	exp education, graduate/ or education, medical/ or exp education, medical, graduate/ or exp education, medical, undergraduate/ or exp physicians/ or exp students, medical/ or exp students, public health/	#14	283 620
Doctor or resident (Free text)	(allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical	#15	1 265 235

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ti. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ab. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).kf.		
Doctor or resident	14 or 15	#16	1 366 539
Measurement properties (Controlled vocabulary)	exp discriminant analysis/ or exp Health Status Indicators/ or exp observer variation/ or exp "Outcome Assessment (Health Care)"/ or exp Psychometrics/ or exp "reproducibility of results"/	#17	1 655 441

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Measurement properties (Free text)	(Instrumentation or methods).fs.	#18	3 941 530
	("Comparative Study" or "Validation Studies").pt.	#19	1 915 070
	(clinimetr* or clinometr* or "outcome measure*").af.	#20	213 666
	(agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ti. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ab. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-	#21	2 263 439

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).kf.		
	(cronbach* and alpha$).ti. or (cronbach* and alpha$).ab. or (cronbach* and alpha$).kf.	#22	20 943
	(item* and (correlation* or selection* or reduction*)).ti. or (item* and (correlation* or selection* or reduction*)).ab. or (item* and (correlation* or selection* or reduction*)).kf.	#23	34 264
	((replicab* or repeated) and (measure$ or finding or result$ or test$)).ti. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).ab. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).kf.	#24	196 579
	(test and retest).ti. or (test and retest).ab. or (test and retest).kf.	#25	24 833
	18 or 19 or 20 or 21 or 22 or 23 or 24 or 25	#26	7 122 366
Measurement properties	17 or 26	#27	7 786 293
Final Total	5 and 10 and 13 and 16 and 27	#28	2 091

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	1
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	1
Support			
Sources	#5a	Indicate sources of financial or other support for the review	1
Sponsor	#5b	Provide name for the review funder and / or sponsor	1
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	1
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	8

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

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BMJ Open

Evaluating communication with parents in paediatric patient-encounters – A Systematic review protocol

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Keywords:	MEDICAL EDUCATION & TRAINING, PAEDIATRICS, Quality in health care < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Evaluating communication with parents in paediatric patient-encounters – A Systematic review protocol

Original article

Registration: Prospero trial # CRD42020151642

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Abstract

Introduction:

Breaking bad news and dealing with difficult patient encounters is a skill that medical residents must learn during their curriculum. Many different tools are available to measure communication quality, but their development and validation processes are often missing. In this paper, we present the protocol of a systematic review aiming to identify the validated tools for measuring communication skills or communication effectiveness with parents in a paediatrics setting in general, including for difficult patient encounters.

Methods and analysis:

We will conduct our systematic review in accordance with the methodology suggested by COSMIN and will report this paper following the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P). We will include the studies in which authors developed and/or validated tools for assessing the quality of communication with families by residents and/or physicians during patient encounters in paediatric settings. Studies assessing communication in telemedicine and studies that use the tool to measure a different outcome than its validation will be excluded. Our search strategy will be developed by a scientific librarian and validated using the PRESS tool. Two reviewers will independently screen the studies for selection, extract data of the ones included, and assess their level of risk of bias using the COSMIN Risk of Bias checklist. We will perform a narrative synthesis on the study selection process, the characteristics of studies and study population, the characteristics of tools identified, their process of development and/or validation, and their psychometric properties. If sufficient data are available, we will do quantitative analyses for each psychometric property.

Ethics and dissemination:

Approval from an ethics committee is not required, as there is no primary data collection. Our findings will be disseminated through a peer-reviewed publication and at local, national and international conferences.

PROSPERO registration number: CRD42020151642

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Strengths

- This systematic review will provide not only a list of tools for evaluating communication with parents in paediatric patient-encounters, but also their validation proofs.

- The search strategy will be revised by a second scientific librarian using the PRESS tool.

- Appropriate guidelines, COSMIN, will be used for conducting the present systematic review.

Limitations

- Grey literature will not be explored.

- Interviews taking place in a telemedicine setting will not be explored for inclusion.

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Introduction

As medical technologies evolve at an accelerating rate, it would be easy to ignore or forget some of the historically fundamental pillars of the medical profession. Among those bases, the patient-physician interaction is a key aspect that requires the continued development of skills, and this appears to be overlooked in some medical faculties [1-3]. To many physicians and residents, difficult patient encounters are a source of anxiety, and they find their approach leaves room for improvement [3, 4]. In paediatrics, communication can be especially challenging because of the patient-parent-physician triad that is present at almost every consultation [5, 6]. Moreover, the way bad news is delivered can influence subsequent patient-parent-physician interactions and the parents' perception, coping capability and acceptance of the disease, as well as affect patient outcomes [3, 5, 7, 8]. Not a lot of literature exists regarding communication between physician and paediatric patients as the child's opinion regarding his doctor's communication skills is less reliable and harder to evaluate with a validated tool [9]. In contrast, many studies focus interest on communication with the patient's parents. As is the case for many skills, a physician's ability to effectively interact with families must be learned and improved during medical training [1, 5, 8, 10, 11].

There are many aspects to consider when assessing difficult patient encounters. The main aspects often cited are those relating to the patient and parent's reality, those relating to the patient's disease, those relating to the medical team's capabilities and those relating to the environment where the encounter takes place. It seems obvious to state that parental anxiety, understanding of the situation and personality will influence the parent-patient-physician relation. Moreover, a complicated medical situation (in terms of severity of illness, prognosis, treatment options) will influence the encounter and add tension to the physician's relation with parents. The physician's experience, personality and workload will also influence the encounter. Environmental factors, such as the healthcare system's organization and the hospital or clinic's facilities can influence the way news will be received by a parent or given by the physician [6]. Those are only some examples of the multitude of factors that can influence difficult patient and parent encounters.

Many medical faculties have taken a recent interest in improving their curriculum's ability to learn communication skills in difficult patient encounters and there are a growing number of tools in literature that are offered to help that achievement by residents. Many residents report high levels of anxiety and low levels of confidence in their communication skills for breaking bad news [4]. In the paediatric objectives of training of the Royal College of Physicians and Surgeons of Canada, some emphasis is made on the importance of good communication as an acquired skill [12]. There is currently a reform in Canada for residency programs to include a competency by design approach to learning, and residents will be specifically evaluated on their communication skills. No official curriculum is however currently available for paediatric residents.

Communication quality is somewhat subjective as many aspects cannot be quantified, and every patient's needs are different [7]. In that context, developing an assessment tool that could accurately measure communication skills is difficult. An ideal the assessment tool should consider the clinical setting. For example, delivering bad news in the emergency ward is not the same as doing so in a palliative care unit. Many tools have been developed to evaluate communication in

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different clinical settings, but there appears to be a lack of validation for these tools [13]. To our knowledge, no comprehensive and systematic review has identified the validated tools for measuring communication skills or communication effectiveness with parents in a paediatrics setting in general, or for difficult patient encounters. Identifying which validated tools exist and have been tested in paediatrics settings would prove very useful to better assess residents' needs regarding their learning of key communication skills for parent-physician interactions and eventually, better address these needs. We took specific interest in difficult patient encounters, because we strongly felt that it is in these situations that communication is more challenging and would benefit the most from better expertise and related training.

Objectives

The present review aims to identify the validated tools for measuring communication skills or communication effectiveness with parents in a paediatric setting in general, including for difficult patient encounters.

Methods

We will conduct a systematic review according to the methodology suggested in COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) guidelines [14-16]. We reported the protocol of our review according to the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-P) [17]. This protocol was registered on PROSPERO (CRD42020151642).

Eligibility criteria

We defined the eligibility criteria of studies using four key elements recommended in the COSMIN Methodology: the construct; the population(s); the type of instrument(s); and the measurement properties of interest. We will also consider study designs and the settings in which the studies are conducted.

1) Construct

The construct of interest will be the evaluation of communication quality in paediatric doctor-parent encounters. We will include doctor-parent encounters for which review of the interaction will be evaluated. Many aspects of communication can be evaluated: the words used, the pace of the interview, the body language are some important aspects that we think will be identified in this review. To be included, the authors will have to present a tool that evaluates communication between a physician and the family of a paediatric patient.

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2) Population(s)

The target population will be doctors of any medical field interacting with paediatric patients' parents or guardians. We believe that a tool evaluating a certified doctor's communication would be applicable to a resident as well and therefore, we will not limit ourselves to studies focused on medical residents. Studies focusing solely on communication between patients and other medical professionals will be excluded. Evaluation of medical student's interactions with parents will be excluded as their learning objectives and expectations differ from those of a resident. If the study includes adult patients, the tool's validation analysis will need to be done at least with a paediatric subgroup (with a patient sample of at least 80% of participants under 16 years old). We chose 16 years old because we believe that beyond that age, the adolescent's parents are less often present during consultations and communication quality assessments should be focused on the adolescent himself and not his parents.

3) Type of instrument(s)

No restriction will apply as to regarding the format of instrument studied. We expect to find many different types of instruments. We expect to find some that are administered by peer review, by the parents themselves or by self-evaluation. Some will be filled in electronic format, others in paper format. In the electronic format, we expect different means of completion will be used: fillable electronic documents, application-based documents, web-based documents. We will only consider instruments which evaluate the construct of interest as a main goal. Therefore, tools that measure general satisfaction following a hospitalization or that measure communication quality with the medical team as a whole and not just doctors or residents specifically will be excluded.

4) Measurement properties

The measurement properties of interest will be any psychometric parameter as defined in the COSMIN taxonomy [18]. Only studies reporting a tool with at least one validation proof or fidelity assessment will be included. The proofs of validation will have to be reported or referenced through other sources in the studies.

5) Study designs

The study designs considered will be development and/or validation studies. We will exclude studies in which authors used the tool for the outcome measurement without its process of development and/or validation as suggested by the COSMIN guidelines [14].

6) Settings

The settings considered will be anywhere physicians or residents interact with parents during a medical encounter. We will exclude interactions taking place in a telehealth context (encounters over the phone, over web platforms, over videoconference, etc..).

As cited earlier, many aspects of communication must be taken into account when considering difficult patient encounters. We believe it would be impossible to address all those aspects in our search strategy. Although articles that analyzed clinical situations that did not represent an aspect of difficulty (for example, that analyzed everyday communication with a patient's family) were not an initial focus of our study, we will not exclude them during the selection process and to instead consider the setting in which the tool was used in the analysis. This will allow us to reduce

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the possible selection bias of having a search strategy not constructed precisely for difficult patient encounters or having to use our judgement to decide which studies took place in a difficult patient encounter setting. The only restrictive criterion regarding the setting will be that the encounter should take place in person. As previously mentioned, medical encounters via electronic messaging, phone or other indirect channels present some specific considerations, and we will not be analyzing them. This protocol was first conceptualized before the COVID-19 pandemic and at that time, telemedicine was not widely used in practice, although it is now expected to gain significant importance. We believe we would find little literature regarding the evaluation of communication in a paediatric setting via telemedicine, and that many criteria for measuring communication quality in face-to-face interviews will be applicable for telemedicine.

Information sources

We will use bibliographic databases such as Medline, Embase, Web of Science, CINAHL, PsycINFO, ERIC-EBSCO and the Cochrane Library. We chose a wide range of databases to cover comprehensive literature on what is available regarding the evaluation of communication in the paediatric setting. We will also consult the lists of bibliographic references in existing relevant reviews (if we find any), and in studies that will be included in the present review.

Search strategy

We developed the preliminary search strategy in Medline with the help of a scientific librarian. We used the following main concepts: communication with families, tools for assessing, paediatric, doctor or resident, and measurement properties. Free and controlled vocabularies of these concepts were combined providing the first version of search strategy. The results of this latter were discussed with team members and their comments were integrated providing a new version. This latter was submitted to another scientific librarian for a revision using the PRESS peer review of electronic search strategies [19]. The final version of search strategy was approved by team members (see supplemental material). We will translate the final version of search strategy in other databases mentioned above.

No language restriction will be applied in the search strategy. If a study seems to be of interest to our review and the full text is not available in French nor English, we will be unable to read it entirely and will mention its omission in our data synthesis.

Grey literature

After consideration, grey literature will not be explored during this review. It certainly would be interesting to explore tools developed that may not have gotten published but doing so in a systematic manner would require tremendous resources and time. There may exist institutions that have created personalized tools that evaluate our construct of interest. Even if these tools have not been validated rigorously, the fact that they may have been used for a long time could serve as validation proofs. However, an environmental scan, including a literature search and/or surveys [20] would have been required to exhaustively identify these tools not published in peer-reviewed journals, as well as their proofs of use. Finally, we believe that most tools that have gone through a rigorous validation process were created with the ultimate objective of being published.

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Therefore, we have limited the present review to the tools published in peer-reviewed journals making irrelevant the grey literature.

Study records

Data management

We will export the citations identified from bibliographic databases to EndNote file. The citations will be merged, and duplicates will be removed to obtain unique citations for the selection of studies. The file will then be exported on Rayyan, a web app designed to facilitate the selection process in systematic reviews [21].

Selection process

We will perform the process of study selection in three steps. Step 1: A pilot will independently be done by the two reviewers (OC and JM) based on 10% of unique citations. This pilot will permit the reviewers to have a common understanding of eligibility criteria. Step 2: After a conclusive pilot, OC and JM will independently perform the selection of studies by title and abstract to exclude early non-eligible articles. Every study selected by one or both reviewers will then be considered for the next step. Step 3: OC and JM will independently select the studies retained at the Step 2 based on the full texts and will document the reasons of exclusion. A third person (CC) will be consulted in case of discordance or uncertainty. We will also contact, through emails, the study authors when information is not clear or missing.

Data collection process

We will develop a codebook and a grid for our data extraction based on the COSMIN methodology [16]. The codebook will include our variables of interest, their definitions, modalities to extract, and specific comments for particular cases. The grid including the variables and their modalities will be used for data extraction. The codebook and grid developed will be independently tested on two articles by two reviewers (OC and JM) to ensure a common understanding. After the conclusive pilot, OC and JM will independently extract the data from included articles. Disagreements observed after data extraction will be discussed by OC and JM for consensus. When the consensus is not reached, a third person (CC) will be consulted for a final decision.

Data items

We will extract data items characterizing the construct, the population, the type of instrument, and the measurement properties of interest. We will also collect information on the development and validation processes and the characteristics of studies.

Here are some of the main elements that will be extracted, when available:

- ***Study characteristics*** **-** name of first author, year of publication, setting, study design
- ***Construct characteristics*** - name, definition, content, number of items, range scores, time to complete, ease of use, if and how feedback is provided to the evaluated doctor
- ***Target population characteristics*** **-** profiles, mean age in years (of the patient and his parent), patient's age group (infant, toddler, preschooler, adolescent), male/female ratio;

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- ***Physician's characteristics*** – mean age in years, experience, specialty, male/female ratio
- ***Setting of the encounter*** – degree of urgency, amount of people in the interview, time of the interview, medical field concerned (emergency, hospital wards, general clinics, specialty clinics)
- ***Type of instruments*** - name, format, administrative type, total number of items, subjective status
- ***Development process*** - development steps, profiles of people involved, statistical methods used
- ***Validation process*** - validation steps, validation type, profiles of participants, mean age of participants, percentage of women participants, statistical methods used, etc.
- ***Psychometric properties*** - reliability (type, name of parameter, value of parameters, p-value, etc.), validity (type, name of parameter, value of parameters, p-value, etc.), responsiveness (type, name of parameter, value of parameters, p-value, etc.), interpretability (type, name of parameter, value of parameters, p-value, etc.).

Risk of bias in individual studies

After a conclusive pilot for the assessment of study quality, OC and JM will independently evaluate the quality of included studies using the COSMIN Risk of Bias checklist [15]. A four-point rating system will be used to grade the methodological quality of each study (very good, adequate, doubtful, inadequate). Content validity, structural validity, reliability, measurement error and other pertinent elements will be evaluated with a rating system with the use of the checklist – a total of 10 boxes will be verified [15]. Each study will be graded according to its lowest-rating methodological aspect. OC and JM will discuss the discordances or uncertainties and will consult CC for the final decision if a consensus is not reached.

We will also apply the updated criteria for good measurement properties for the classification of identified tools [14]. Each of the eight criteria will be rated sufficient, insufficient or indeterminate according to the standards described in the COSMIN methodology [14].

Synthesis

We will describe the selection process using the PRISMA flowchart and frequency counts. We will determine the concordance between reviewers at the steps 2 and 3 of study selection calculating a Cohen's kappa coefficient. The concordance will be considered sufficient if the kappa is equal or greater than 0.70 [22].

We will perform a narrative synthesis using descriptive statistics. Characteristics of studies, construct, population, instrument, and measurement properties will be analyzed using frequency counts for dichotomous or categorical variables and means, medians and interquartile ranges for continuous variables. We will also analyze the data on the study quality assessment, the tool development and validation processes using graphic and frequency counts. We will especially be interested in looking at the context and the population characteristics the tool was used in, including the age of the patients, which criteria the tool used to assess communication quality and how was the tool validated.

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If we obtain sufficient data on parameters of psychometric properties, we will perform a quantitative analysis on each of them following the methods suggested in COSMIN guideline [14].

Meta-bias

Because of the nature of our review, we will not assess meta-biases. As is stated in the COSMIN methodology, publication bias is difficult to assess in systematic reviews of patient-reported outcome measures (PROMs) because of a lack of registries for studies on measurement properties [14].

Confidence in cumulative evidence

The strength of the body of evidence will be assessed using the modified grading of recommendation assessment, development and evaluation (GRADE). As proposed in the COSMIN methodology [16], we will use four of the five GRADE criteria: risk of bias, i.e., the methodological quality of the included studies; inconsistency, i.e., unexplained inconsistency of results across included studies; imprecision, i.e., total sample size of the available included studies and indirectness, i.e., evidence from different populations than the population of interest in the review [14]. The quality of the evidence will be graded for each measurement property separately. The starting point will always be the assumption that the pooled or overall measurement criteria is of high quality. The quality of evidence for each measurement property subsequently downgraded by one or two levels per criterion to moderate, low, or very low [14].

Ethics and dissemination

No ethic approbation is necessary as there is no primary data collection. Our findings will be disseminated through a peer-review publication, at local, national and international conferences. It will also be disseminated to the team's affiliated University for possible improvement of the curriculum. This protocol was registered with PROSPERO.

Patient and public involvement

Patients or the public were not involved in the design of this study. The results of this study will be disseminated through a peer-review publication and at local, national and international conferences, which will not necessarily be open to public. As no primary data collection is required, patient involvement's added value is assessed as minimal in this context.

Discussion:

We believe that this review's methodological quality will be high because of the different choices we have made along the elaboration of this protocol. Every step of conceptualization of this review was performed in accordance with the PRISMA and COSMIN guidelines [14-17]. The search strategy was developed by a team of qualified researchers and revised by an independent librarian with the PRESS review [19]. The selection process and analysis will be done rigorously by two independent reviewers. We will unfortunately not assess meta biases and search for grey literature during this review, which is a limitation.

This review was elaborated during the COVID-19 pandemic and will not be addressing encounters over a telemedicine setting. We believe that literature addressing telemedicine is prone to expand

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greatly in the next couple of years as an increase in usage of telemedicine technologies is currently seen all around the world. It would be interesting, as a subsequent study, to review the literature analysing the evaluation of communication quality over a telemedicine setting. This review's methodological construct and content will regardless be of use to whoever may take interest in this topic.

We believe this systematic review will serve as a portrait of the available evaluation tools with their validation proofs that measure the quality of communication in paediatric encounters. It will be available to use for physicians and researchers who want to use those tools in a trial, to educators who want to further their teaching abilities and to faculties or institutions who aim at evaluating adequately their residents' communication skills. Medical schools' curriculums could benefit by having a better understanding of their resident's communication skills during their paediatric rotations and potentially a better idea of precise aspects they need to work with a resident to improve those skills. We believe this review could set guidance in which ways communication should be evaluated by the Royal College of Physicians and Surgeons of Canada, specifically in paediatrics. Furthermore, the ultimate goal is providing physicians the best tools to improve their communication with families, especially during difficult patient encounters, thus improving parent satisfaction. We believe having the highest quality of interactions with families can and will improve patients' quality of care and satisfaction with clinical encounters.

Contributorship statement

Olivier Cusson (Corresponding Author): Substantial contribution to all stages of the review, approved the final version and agrees to be accountable for its content.

Justine Mercier: Substantial contribution to the interpretation of the data for the review, revision of all its versions, approval of the final version and agrees to be accountable for its content.

Céline Catelin: Substantial contribution to the conception of the review, revision of all versions of the protocol, approval of the final version and agrees to be accountable for its content.

Hervé Tchala Vignon Zomahoun: Substantial contribution to the design of the review, revision of all its versions, approval of the final version and agrees to be accountable for its content.

Acknowledgements

Nathalie Rheault: Search strategy conceptualization

Rachel Couban: PRESS review of the search strategy

Marie-Claude Battista: Initial methodology expert

Competing interests

None

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University of Sherbrooke – Department of Paediatrics (no grant number)

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Data sharing statement

Data generated by this article will be made openly and publicly available upon publication.

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Supplemental material:

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Communication with families (Controlled vocabulary)	exp Hospital-Patient Relations/ or exp Physician-Patient Relations/	#1	72 086
Communication with families (Free text)	((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ti. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).ab. or ((communicat* or conversat* or discuss* or empower* or encounter* or interact* or interview* or participation or relation*) adj4 (caregiver* or "care giver*" or caretaker* or client* or family or families or father? or guardian* or mother? or parent* or patient? or people or person? or tutor?)).kf.	#2	268 858
	("non technical" adj3 (abilit* or competenc* or skill?)).ti. or ("non technical" adj3 (abilit* or competenc* or skill?)).ab. or ("non technical" adj3 (abilit* or competenc* or skill?)).kf.	#3	650
	2 or 3	#4	269 475
Communication with families	1 or 4	#5	321 467
Tools for assessing (Controlled vocabulary)	exp Checklist/ or exp guideline/ or "Surveys and Questionnaires"/	#6	470 373

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Tools for assessing (Free text)	((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ti. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).ab. or ((checklist$ or "check list$" or guide$ or guideline$ or instrument* or questionnaire$ or scale$ or survey$ or tool*) adj4 (analy* or assess* or evaluat* or measur* or observation* or observe or observing or track*)).kf.	#7	447 741
	("assessment method*" or "evaluation method*").ti. or ("assessment method*" or "evaluation method*").ab. or ("assessment method*" or "evaluation method*").kf.	#8	17 715
Tools for assessing (Free text)	7 or 8	#9	462 404
Tools for assessing	6 or 9	#10	844 912
Pediatric (Controlled Vocabulary)	exp adolescent/ or exp child/ or exp Hospitals, Pediatric/ or exp infant/ or exp Pediatric Emergency Medicine/ or exp Pediatric Nursing/ or exp Pediatricians/	#11	3 442 520
Pediatric (Free text)	(adolescen* or child or childhood or children or infan* or p?ediatric*).ti. or (adolescen* or child or childhood or children or infan* or p?ediatric*).ab. or (adolescen* or child or childhood or children or infan* or p?ediatric*).kf.	#12	1 860 446
Pediatric	11 or 12	#13	3 866 495
Doctor or resident (Controlled vocabulary)	exp education, graduate/ or education, medical/ or exp education, medical, graduate/ or exp education, medical, undergraduate/ or exp physicians/ or exp students, medical/ or exp students, public health/	#14	283 620
Doctor or resident (Free text)	(allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical	#15	1 265 235

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ti. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).ab. or (allergist* or anesthesiologist* or cardiologist* or clinician* or dermatologist* or doctor$ or endocrinologist* or gastroenterologist* or "general practitioner*" or gps or graduate* or hospitalist* or intern or interns or "medical education" or "medical facult*" or "medical school*" or neonatologist* or nephrologist* or neurologist* or oncologist* or ophthalmologist* or orthop?edist* or otolaryngologist* or p?ediatrician* or physiatrist* or physician* or podiatrist* or pulmonologist* or radiologist* or rheumatologist* or resident$ or "medical student*" or "clinical student*" or "preclinical student*" or surgeon* or trainee? or undergraduate* or urologist*).kf.		
Doctor or resident	14 or 15	#16	1 366 539
Measurement properties (Controlled vocabulary)	exp discriminant analysis/ or exp Health Status Indicators/ or exp observer variation/ or exp "Outcome Assessment (Health Care)"/ or exp Psychometrics/ or exp "reproducibility of results"/	#17	1 655 441

Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
Measurement properties (Free text)	(Instrumentation or methods).fs.	#18	3 941 530
	("Comparative Study" or "Validation Studies").pt.	#19	1 915 070
	(clinimetr* or clinometr* or "outcome measure*").af.	#20	213 666
	(agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ti. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).ab. or (agreement* or coefficient* or "discriminant analys*" or "health status indicator*" or homogeneity or homogeneous or imprecision* or interassay or "inter-assay" or interexaminer* or "inter-examiner*" or interindividual* or "inter-	#21	2 263 439

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Literature search strategy in Ovid Medline (2019-08-27)			
Concepts	**Search strategy**	**Number**	**# Results**
	individual*" or "internal consistenc*" or intertechnician* or "inter-technician*" or interobserver* or "inter-observer*" or interparticipant* or "inter-participant*" or interrater* or "inter-rater*" or intertester* or "inter-tester*" or intraassay* or "intra-assay*" or intraexaminer* or "intra-examiner*" or intraindividual* or "intra-individual*" or intraobserver* or intraparticipant* or "intra-participant*" or intrarater* or "intra-rater*" or intratechnician* or "intra-technician*" or intratester* or "intra-tester*" or kappa$ or "observer variation*" or "outcome? assessment" or "precise value$" or precision* or psychometr* or repeatab* or reproducib* or reliab* or stability or "test-retest" or unreliab* or valid*).kf.		
	(cronbach* and alpha$).ti. or (cronbach* and alpha$).ab. or (cronbach* and alpha$).kf.	#22	20 943
	(item* and (correlation* or selection* or reduction*)).ti. or (item* and (correlation* or selection* or reduction*)).ab. or (item* and (correlation* or selection* or reduction*)).kf.	#23	34 264
	((replicab* or repeated) and (measure$ or finding or result$ or test$)).ti. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).ab. or ((replicab* or repeated) and (measure$ or finding or result$ or test$)).kf.	#24	196 579
	(test and retest).ti. or (test and retest).ab. or (test and retest).kf.	#25	24 833
	18 or 19 or 20 or 21 or 22 or 23 or 24 or 25	#26	7 122 366
Measurement properties	17 or 26	#27	7 786 293
Final Total	5 and 10 and 13 and 16 and 27	#28	2 091

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	1
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	1
Support			
Sources	#5a	Indicate sources of financial or other support for the review	1
Sponsor	#5b	Provide name for the review funder and / or sponsor	1
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	1
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	8

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

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